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4次作业， Homework 20%. Exam 80%

Real Analysis → <<数学分析>> 华东师范大学（第5版）

1. 基础：记号，集合，区间，距离
2. 函数与极限： ε-δ 定义，极限存在性，夹逼定理，极限运算
3. 连续函数： 连续性定义，有界性，最值定理，介值定理，
   一致连续
4. 求导： 求法则，链式法则，隐函数，
   罗尔定理，拉格朗日中值定理，洛必达法则，
   泰勒定理。

Notation

Space： $\mathbb{R}$：the set of real numbers

$\mathbb{Q}$：the set of rational numbers ratio

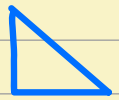
$x \in \mathbb{Q} \Longleftrightarrow \{ x = \frac{p}{q},\ q \neq 0,\ p, q \in \mathbb{Z} \}$

$\mathbb{Z}$：the set of integal numbers.

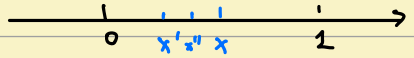
毕达哥拉斯学派 → 希帕索斯

↓ ↓

$a^2 + b^2 = c^2$ $1^2 + 1^2 = 2$



$\mathbb{R} = \mathbb{Q} \cup \{$ the set of irrational number $\}$ 无理数



稠密的

Fact：

The set of rational numbers is dense

The set of real numbers is continuous

连续的

KBC 定义

9个 claim

1.1. Set theory

非正式定义

A set is a collection of different things.

Properties (性质)

① well-definedness (确定性) either $a \in A$ (in) or $a \notin A$ (not in)

② Distinct elements (互异性) $A = \{1, 6, 3\}$ $B = \{1, 3, 6, 6\}$

③ Unorder (无序性) $A = B$ $= \{1, 3, 6\}$

Notations: $A \subset B \Leftrightarrow \{A \subseteq B \text{ and } A \neq B\}$

Property: If $A \subseteq B$, and $B \subseteq A$, if and only if $A = B$.

Union (并集) $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Intersection (交集) $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Complement (补集) $A^c = \{x \mid x \notin A\}$

Empty set: $\emptyset$ a set that contains no elements.

罗素悖论: 例1

$S = \{x \mid x \notin S\}$ $S \in S$?

Case 1: If $S \in S$, by definition, $S \notin S$.

case 2: If $S \notin S$, by definition, $S \in S$.

数学的公理化

Index sets: $A_1, A_2, A_3, \cdots A_n \rightarrow$ sets

$I = \{1, 2, \cdots, n\} \rightarrow$ Index sets.

$A_1 \cup A_2 \cup A_3 \cdots \cup A_n =: \bigcup_{i \in I} A_i = \{x \mid x \in A_i \text{ for some } i \in I\}$. (cup)

$A_1 \cap A_2 \cdots \cap A_n =: \bigcap_{i \in I} A_i = \{x \mid x \in A_i \text{ for all } i \in I\}$ (cap)

De Morgan's laws:

① $\left(\bigcup_{i\in I} A_i\right)^c = \bigcap_{i\in I} A_i^c$

② $\left(\bigcap_{i\in I} A_i\right)^c = \bigcup_{i\in I} A_i^c$

1.2 The Intervals (区间)

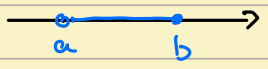
Defn: An interval is a subset of the real numbers $\mathbb{R}$. given $a, b \in \mathbb{R}$, $a < b$

$]a,b[\ \overset{\text{记}}{=} (a,b)$

Notations:

Finite interval:

- $(a,b) := \{x \in \mathbb{R} \mid a < x < b\}$ open interval
- $[a,b] := \{x \in \mathbb{R} \mid a \le x \le b\}$ close interval.
- $[a,b) := \{x \in \mathbb{R} \mid a \le x < b\}$
- $(a,b] := \{x \in \mathbb{R} \mid a < x \le b\}$



Infinite interval

- $(a, +\infty) := \{x \in \mathbb{R} \mid x > a\}$
- $[a, +\infty) := \{x \in \mathbb{R} \mid x \ge a\}$
- $(-\infty, b) := \{x \in \mathbb{R} \mid x < b\}$
- $(-\infty, b] := \{x \in \mathbb{R} \mid x \le b\}$.

Notation: $\mathbb{R}_+ := [0, +\infty)$ $\mathbb{R}_{++} := (0, +\infty)$

E.g.: Given $x, y \in \mathbb{R}$, $x < y$. then there exists a rational number r such that

$$x < r < y.$$

Proof: Since $x < y$, there exists $n \ge 0$, such that $\overline{x}_n < \underline{y}_n$. Let $r = \frac{1}{2}(\overline{x}_n + \underline{y}_n)$

$\Rightarrow x \le \overline{x}_n < r < \underline{y}_n < y$

we conclude $x < r < y$, and r is a rational number.

$y = 11.346$

$x = 11.346165\ldots$

$a_0 = 11$ $a_1 = 3$, $a_2 = 4$

$a_3 = 7$ $a_4 = 1$

$x = a_0.a_1a_2a_3a_4\ldots$

过剩近似: $\overline{x}_n = 11.3472$ $n = 4$

不足近似: $\underline{x}_n = 11.3461$ $n = 4$

$\underline{x}_n \le x \le \overline{x}_n$

Properties of $\mathbb{R}$

$(+, -, \times, \div)$

① closure property (封闭性) Any arithmetic operation is performed on two elements of $\mathbb{R}$, with the answer being another element of $\mathbb{R}$.

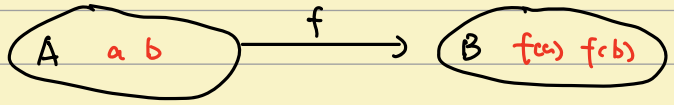
② Ordered (有序性) any two elements of $\mathbb{R}$, one of the following holds: $a<b$, $b<a$, $a=b$.

③ Transitivity (传递性) if $a<b$, $b<c \Leftrightarrow a<c$

④ Density (稠密性) if $a<b$, then there exists $c \in \mathbb{R}$, such that $a<c<b$.

single-valued function →

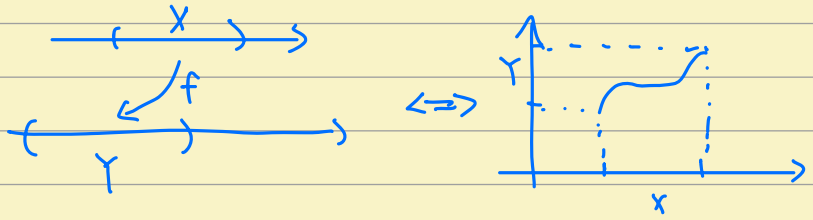
1.3 Functions (函数)



Real function $f: X \to Y$ with $X, Y \subseteq \mathbb{R}$

X: the domain (定义域) of f: $\mathrm{Dom}(f)$

Y: the range / image (值域) of f: $\mathrm{Im}(f)$ $\mathrm{Range}(f)$



Graph (图) of f: $\{(x, y) \in \mathbb{R}^2 \mid y = f(x),\ x \in X,\ y \in Y\}$

E.g.: $f(x) = 1$ $x \in \mathbb{R}$ $\mathrm{Dom}(f): \mathbb{R}$, $\mathrm{Im}(f): \{1\}$

$f(x) = |x|$ $x \in \mathbb{R}$ $\mathrm{Dom}(f): \mathbb{R}$, $\mathrm{Im}(f): \mathbb{R}_+$

$f(x) = \sqrt{x^2}$ $x \in \mathbb{R}_+$ $\mathrm{Dom}(f): \mathbb{R}_+$, $\mathrm{Im}(f): \mathbb{R}_+$

Defn: Let $f: X \to Y$ be a real function

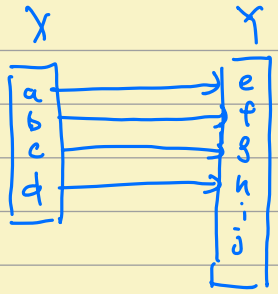
① The function is injective (单射) (or one-to-one) if for any $a, b \in X$, $f(a) = f(b)$ implies $a = b$

The function is called an injection (单射函数)

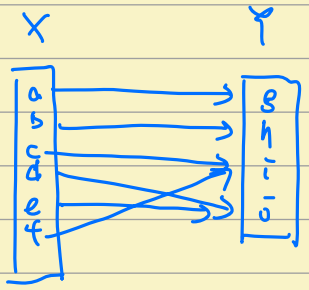
② The function is surjective (满射) if $f(X) = Y$

The function is called a surjection (满射函数)

Injection



Surjection



E.g.:

$f(x) = e^x$	$f: \mathbb{R} \to \mathbb{R}$	injective but not surjective
$f(x) = x^2$	$f: \mathbb{R} \to \mathbb{R}_+$	surjective but not injective
$f(x) = x^2$	$f: \mathbb{R} \to \mathbb{R}$	neither injective nor surjective
$f(x) = x^3$	$f: [-1, 1] \to [-1, 1]$	both injective and surjective

③ The function f is bijective (双射) if it is both injective and surjective. The function is call bijection.

Operations of the functions

Given real functions : $f_1 : X_1 \to Y_1$, $f_2 : X_2 \to Y_2$. $X := X_1 \cap X_2$ if $X \neq \phi$.

$$\left.\begin{aligned} F(x) &= f_1(x) + f_2(x) \\ G(x) &= f_1(x) - f_2(x) \\ H(x) &= f_1(x) \cdot f_2(x) \end{aligned}\right\} \Rightarrow \quad dom(\cdot) = X$$

$$L(x) = \frac{f_1(x)}{f_2(x)} \Rightarrow dom(L) = X_1 \cap \{x \in \mathbb{R} \mid f_2(x) \neq 0 \quad x \in X_2\}$$

E.g. $f_1(x) = \sqrt{1-x^2}$ $f_2(x) = \sqrt{x^2-9}$

$F(x) = f_1(x) + f_2(x)$.

If a function f is not well-defined if $dom(f) = \phi$.

$X = X_1 \cap X_2$ $X_1 = [0, 1]$ $X_2 = [2, +\infty)$

$X = \phi$ $\therefore$ $F(x)$ is not well-defined.

Composed function (合成函數)

Given $y = f(u)$ $u \in D$, $u = g(x)$, $x \in X$.

Let $X' = X \cap \{x \in \mathbb{R} \mid g(x) \in D\} \neq \phi$, then the composed function is denoted as

$$y = f(g(x)) = f \circ g(x)$$

E.g.: $y = f(u) = \sqrt{u}$ $u = g(x) = 1 - x^2$ $X = \mathbb{R}$

$y = f \circ g(x) = \sqrt{1-x^2}$

$X' = X \cap \{x \in \mathbb{R} \mid g(x) \in D\} = \mathbb{R} \cap [-1, 1] = [-1, 1]$

Inversed function (反函数)

Given a real function : $f: X \to Y$, its inversed function is

$$f^{-1}: f(X) \to X$$

E.g.: $f(x) = \frac{1}{2}x$ $\quad f: [-1, 1] \to [-1, 1]$

$f^{-1} = 2x$ $\quad f^{-1}: [-\frac{1}{2}, \frac{1}{2}] \to [-1, 1]$

Monotonic function (单调函数)

Defn: A real function $f: X \to Y$, $\forall x_1, x_2 \in X$, $x_1 < x_2$ if

① $f(x_1) \le f(x_2)$, then f is monotonically increasing

② $f(x_1) < f(x_2)$, then f is strictly monotonically increasing

③ $f(x_1) \ge f(x_2)$, then f is monotonically decreasing

④ $f(x_1) > f(x_2)$, then f is strictly monotonically decreasing

E.g.: $f(x) = C$ both m- increasing and m- decreasing.

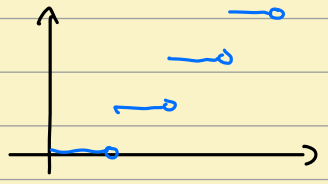
$f(x) = x^2$ $\quad f: \mathbb{R} \to \mathbb{R}_+$ neither

$f(x) = x^3$ Let $x_1 < x_2$

$$f(x_1) - f(x_2) = x_1^3 - x_2^3 = (x_1 - x_2)[x_1^2 + x_1x_2 + x_2^2]$$
$$= (x_1 - x_2)\left[\left(x_1 + \frac{x_2}{2}\right)^2 + \frac{3}{4}x_1^2\right]$$
$$< 0$$

$\Rightarrow f$ is strictly m- increasing.

$f(x) = \lceil x \rceil$



Thm: Let $y = f(x)$, $f: X \to Y$, If f is strictly monotonically increasing, then its inversed function f^{-1} is also strictly monotonically increasing

Thm: decreasing.

1.4 Euclidean distance on the real space (欧氏距离)

Absolute value $|a| = \begin{cases} a & a \geq 0 \\ -a & a < 0 \end{cases}$

Properties:

① $|a| = |-a| \geq 0$, $|a| = 0$ iff $a = 0$

② $-|a| \leq a \leq |a|$

③ $|a| \leq h \iff -h \leq a \leq h$

④ $a, b \in \mathbb{R}$, $|a| - |b| \leq |a \pm b| \leq |a| + |b|$

⑤ $|a \cdot b| = |a| \cdot |b|$

⑥ $\frac{|a|}{|b|} \leq \left|\frac{a}{b}\right|$ $(b \neq 0)$

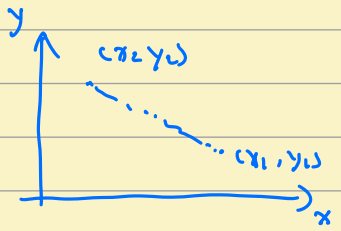
Defn: On $\mathbb{R}$, the Euclidean distance is the function

$\mathbb{R}\times\mathbb{R}$ → 平面欧氏距离

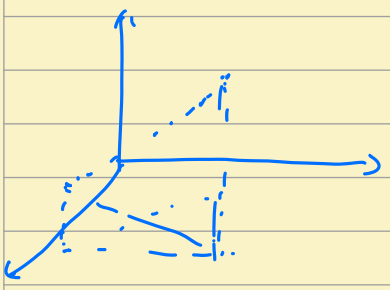
$$d : \mathbb{R}\times\mathbb{R} \to [0, +\infty)$$
$$d(x,y) = |x-y| = \sqrt{(x-y)^2} \quad \forall\, x, y \in \mathbb{R}$$



$$d(x_1, x_2) = |x_1 - x_2|$$



$$d(x_1, x_2) = \sqrt{(x_1-x)^2 + (y_1-y_2)^2}$$



$$d(x_1, x_2) = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2 + (z_1-z_2)^2}$$

Properties of distance function

① $d(x, y) \geq 0$, $d(x, y) = 0$ iff $x = y$ 非负性

② $d(x, y) = d(y, x)$ 对称性

③ $d(x, y) \leq d(x, z) + d(y, z)$ 三角不等式